

Protocol specification



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UDP CONFIGURATION PROTOCOL

Version 3.01

Info: <http://www.duotecno.be>

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TODO:

- Mogelijkheid om Software & protocolversie op te vragen...
- Commando om Devices te resetten – Nodig na uploaden van de app cfg instellingen.

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Introduction:

Packet structure:

All data is formatted in a ASCII command string format. A command string starts via a start character and ends with an end character. All data values are separated via a separator character.

Start command string character.

Fixed character '[' used for synchronization.

Data values:

The data values are represented as decimal byte values (0 - 255). Larger integers which are represented by multiple bytes are always transmitted and interpreted as most significant byte first, followed by the least significant byte.

Separator character:

Fixed character ',' used to separate the data values.

End command string character:

Fixed character ']' used to signal the command string has ended.

End of message character

Fixed character 'LF' (h0a) used to signal the server can process the received command strings. For compatibility reasons 'CR' + 'LF' pair is allowed. (h0d,h0a).

Example:

Command string = [163,2,3,0]

Byte#	Char	Description
0	'['	Start char
1	'1'	Data: Command
2	'6'	Data: Command
3	'3'	Data: Command
4	','	Separator
5	'2'	Data: Method
6	','	Separator
7	'3'	Node Address
8	','	Separator
9	'0'	Unit address
10	']'	End command string
11	LF (h0A)	End character.

Supported communication media:

UDP/IP Protocol

Duotecno devices listen to a port (default port = 5002).

The host configuration software sends address oriented and broadcast LAN messages to this UDP port.

Address oriented messages needs to be implemented as a request/response mechanism.

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Commands overview.

Id	Name	Since protocol:	Changes protocol:	Note
12	EV_NETWORKCONFIG	V1.00		
15	EV_APPCONFIG	V1.00		
184	FC_REQUESTNETWORKCONFIG	V1.00		
185	FC_NETWORKKCONFIG	V1.00		
189	FC_APPCONFIG	V1.00		

EV_NETWORKCONFIG

Object	Since protocol:
0 = OBJECT_FOUND_IPV4	V1.00
1 = ATTR_CFG_IPV4	V1.00

EV_APPCONFIG

Object	Since protocol:
0 = ATTR_HOMESRVCFG_GET	V1.00
1 = ATTR_HOMESRVCFG_SET	V1.00
2 = ATTR_DDNS_GET	V1.00
3 = ATTR_DDNS_DISABLE	V1.00
4 = ATTR_DDNS_ENABLE	V1.00
6 = ATTR_SYSTEMMODE_GET	V1.00
8 = ATTR_TCIPCLIENT_GET	V1.00
10 = ATTR_DNSPROVIDER_GET	V1.00
12 = ATTR_SNTP_GET	V1.00
14 = ATTR_TIME_GET	V1.00

FC_NETWORKCONFIG

Object	Since protocol:
1 = ATTR_CFG_IPV4	V1.00

FC_REQUESTNETWORKCONFIG

Object	Since protocol:
0 = OBJECT_FIND_IPV4	V1.00

FC_APPCONFIG

Object	Since protocol:
0 = ATTR_ENUM_SUPPORTED	V1.00

1 = ATTR_SRVCFG_GET	V1.00
2 = ATTR_SRVCFG_SET	V1.00
3 = ATTR_DDNS_GET	V1.00
4 = ATTR_DDNS_DISABLE	V1.00
5 = ATTR_DDNS_ENABLE	V1.00
6 = ATTR_SYSTEMMODE_GET	V1.00
7 = ATTR_SYSTEMMODE_SET	V1.00
8 = ATTR_TCIPCLIENT_GET	V1.00
9 = ATTR_TCIPCLIENT_SET	V1.00
10 = ATTR_DNSPROVIDER_GET	V1.00
11 = ATTR_DNSPROVIDER_SET	V1.00
13 = ATTR_SNTP_SET	V1.00
15 = ATTR_TIME_SET	V1.00

Network configuration

Overview:

The host software program broadcast a request to all TCP/IP nodes on the LAN network. All nodes reply with the current network configuration.

When the network configuration is changed by the host software the addressed node replies with an ACK/NAK message.

The host software needs to periodic poll the network to get the latest configuration of the nodes.

Packet fields:

[Command , method , data , data N-1]

Messages:

FC_REQUESTNETWORKCONFIG [184]

General:

The host configuration software broadcast this message in the LAN network to search for Duotecno TCP/IP devices. This message is send to the LAN broadcast IP-address.

Message structure:

Frame	Value	Description
Command	184	Command FC_REQUESTNETWORKCONFIG
Method	0	Method: OBJECT_FIND_IPV4
Data 0	0	Reserved: always 0

Example:

[184,0,0]

FC_NETWORKCONFIG [185]

General:

The host configuration software sends this message to the Duotecno device to configure the network. This message is send to the LAN broadcast IP-address.

Message structure:

Frame	Value	Description
Command	185	Command: FC_NETWORKCONFIG
Method	1	Method: ATTR_CFG_IPV4
Data 0	MAC_5	Mac Address – Byte 5
Data 1	MAC_4	Mac Address – Byte 4
Data 2	MAC_3	Mac Address – Byte 3 (High MSB Physical Address)
Data 3	MAC_2	Mac Address – Byte 2 (High LSB Physical Address)
Data 4	MAC_1	Mac Address – Byte 1 (Low MSB Physical Address)

Data 5	MAC_0	Mac Address – Byte 0 (Low LSB Physical Address)
Data 6	DHCP	0 = Off; 1=On
Data 7	IP_Address_3	IP-Address – BYTE 3
Data 8	IP_Address_2	IP-Address – BYTE 2
Data 9	IP_Address_1	IP-Address – BYTE 1
Data 10	IP_Address_0	IP-Address – BYTE 0
Data 11	Netmask_3	Netmask – BYTE 3
Data 12	Netmask_2	Netmask – BYTE 2
Data 13	Netmask_1	Netmask – BYTE 1
Data 14	Netmask_0	Netmask – BYTE 0
Data 15	Gateway_3	Gateway – BYTE 3
Data 16	Gateway_2	Gateway – BYTE 2
Data 17	Gateway_1	Gateway – BYTE 1
Data 18	Gateway_0	Gateway – BYTE 0

EV_NETWORKCONFIG [12]

General:

The devices reply to the FC_REQUESTNETWORKCONFIG and FC_NETWORKCONFIG messages.

Message structure:

Frame	Value	Description
Command	184	Command FC_NETWORKCONFIG
Method	Method 0-1	Method – see table.
Data 0 - N	Data	F(method) – see below

Methods

Method	Value	Description
OBJECT_FOUND_IPV4	0	Reply to the FC_REQUESTNETWORKCONFIG message with the current network config.
ATTR_CFG_IPV4	1	Reply to the FC_NETWORKCONFIG message with result code.

Message structure - OBJECT_FOUND_IPV4

Frame	Value	Description
Command	12	Command: EV_NETWORKCONFIG
Method	0	Method: OBJECT_FOUND_IPV4
Data 0	0	Reserved: always 0
Data 1	Length	Nr of Char in Name.
Data 2 ... N	Device Name	Max 15 char. (Node Name)
Data N+1	MAC_5	Mac Address – Byte 5
Data N+2	MAC_4	Mac Address – Byte 4
Data N+3	MAC_3	Mac Address – Byte 3 (High MSB Physical Address)

Data N+4	MAC_2	Mac Address – Byte 2 (High LSB Physical Address)
Data N+5	MAC_1	Mac Address – Byte 1 (Low MSB Physical Address)
Data N+6	MAC_0	Mac Address – Byte 0 (Low LSB Physical Address)
Data N+7	DHCP	0 = Off; 1=On
Data N+8	IP_Address_3	IP-Address – BYTE 3
Data N+9	IP_Address_2	IP-Address – BYTE 2
Data N+10	IP_Address_1	IP-Address – BYTE 1
Data N+11	IP_Address_0	IP-Address – BYTE 0
Data N+12	Netmask_3	Netmask – BYTE 3
Data N+13	Netmask_2	Netmask – BYTE 2
Data N+14	Netmask_1	Netmask – BYTE 1
Data N+15	Netmask_0	Netmask – BYTE 0
Data N+16	Gateway_3	Gateway – BYTE 3
Data N+17	Gateway_2	Gateway – BYTE 2
Data N+18	Gateway_1	Gateway – BYTE 1
Data N+19	Gateway_0	Gateway – BYTE 0

Example:

TODO

Message structure – ATTR_CFG_IPV4

Frame	Value	Description
Command	12	Command: EV_NETWORKCONFIG
Method	1	Method: OBJECT_FOUND_IPV4
Data 0	0/1	0 = Failed / 1= Success.
Data 1	MAC_5	Mac Address – Byte 5
Data 2	MAC_4	Mac Address – Byte 4
Data 3	MAC_3	Mac Address – Byte 3 (High MSB Physical Address)
Data 4	MAC_2	Mac Address – Byte 2 (High LSB Physical Address)
Data 5	MAC_1	Mac Address – Byte 1 (Low MSB Physical Address)
Data 6	MAC_0	Mac Address – Byte 0 (Low LSB Physical Address)

Example:

TODO

Enumeration of application supported features.

Overview:

This procedure can be used by the host software to know which configuration properties are supported.

Packet fields:

[Command , method , data , data N-1]

Messages:

FC_APPCONFIG [189]

General:

The host configuration software sends this message to the device to request the features the application supports.

Message structure:

Frame	Value	Description
Command	189	Command FC_APPCONFIG
Method	0	ATTR_ENUM_SUPPORTED

Example:

[189,0]

EV_APPCONFIG [15]

General:

The device replies to the host with this message in response of a FC_APPCONFIG message.

Frame	Value	Description
Command	15	Command EV_APPCONFIG
Method	0	ATTR_ENUM_SUPPORTED
Data 0	Data Flags	Bit 24-31 : See table data flags
Data 1	Data Flags	Bit 16 – 23 : See table data flags
Data 2	Data Flags	Bit 8-15 : See table data flags
Data 3	Data Flags	Bit 0-7 : See table data flags

Table – Data flags:

Bit number	Flag
0	SUPPORTED_FLAG_TCIPSERVER
1	SUPPORTED_FLAG_DDNSCLIENT
2	SUPPORTED_FLAG_SYSTEMMODE
3	SUPPORTED_FLAG_TCIPCLIENT
4	SUPPORTED_FLAG_DNSPROVIDER

5	SUPPORTED_FLAG_SNTP
6	SUPPORTED_FLAG_TIME

Examples:

Message	Description
[15,0,0,0,7]	Device supports: App Homeserver, DDNS Client, System mode configuration.
[15,0,0,0,4]	Device only supports app System mode configuration.

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Application TCP/IP server configuration.

Overview:

Request and set the application homeserver tcp/ip settings.

Packet fields:

[Command , method , data , data N-1]

Messages:

FC_APPCONFIG [189]

General:

The host configuration software sends this message to the device to request and set the Application tcp/ip settings.

Message structure:

Frame	Value	Description
Command	189	Command FC_APPCONFIG
Method	0-1	See table Methods

Method	Value	Description
ATTR_SRVCFG_GET	0	Get the current server login settings.
ATTR_SRVCFG_SET	1	Set the server login settings.

Message structure – ATTR_HOMESRVCFG_GET:

Frame	Value	Description
Command	189	Command FC_SETAPPCONFIG
Method	0	Method: ATTR_SRVCFG_GET

Message structure – ATTR_HOMESRVCFG_SET:

Frame	Value	Description
Command	189	Command FC_SETAPPCONFIG
Method	1	Method: ATTR_SRVCFG_SET
Data 0	LEN	Length of password string
Data 1...N	Password	Password string
Data N+1	IP Port	IP port MSB
Data N+2	IP Port	IP port LSB
Data N+3	Flags	Flags MSB
Data N+4	Flags	Flags LSB

EV_APPCONFIG [15]

General:

The device replies this message to the host software in response of a FC_APPCONFIG message.

Frame	Value	Description
Command	15	Command EV_APPCONFIG
Method	Method 0-1	Method – see table.
Data 0 - N	Data	F(method) – see below

Methods

Method	Value	Description
ATTR_SRVCFG_GET	0	Reply with the current settings
ATTR_SRVCFG_SET	1	Reply to the FC_SETAPPCONFIG message with a result code.

Message structure – ATTR_HOMESRVCFG_GET

Frame	Value	Description
Command	15	Command EV_APPCONFIG
Method	0	Method: ATTR_SRVCFG_GET
Data 0	LEN	Length of password string
Data 1...N	Password	Password string
Data N+1	IP Port	IP port MSB
Data N+2	IP Port	IP port LSB
Data N+3	Flags	Flags MSB
Data N+4	Flags	Flags LSB

Example:

Todo:

Message structure – ATTR_HOMESRVCFG_SET

Frame	Value	Description
Command	15	Command EV_APPCONFIG
Method	1	Method: ATTR_SRVCFG_SET
Data 0	0 or 1	0 = Failed / 1= Success.

Example:

Todo:

Application DDNS client configuration.

Overview:

Request and set the application DDNS client settings.

Packet fields:

[Command , method , data , data N-1]

Messages:

FC_APPCONFIG [189]

General:

The host configuration software sends this message to the device to request and set the Application DDNS client settings.

Message structure:

Frame	Value	Description
Command	189	Command FC_APPCONFIG
Method	2-4	See table Methods

Methods

Method	Value	Description
ATTR_DDNS_GET	2	Get the current DDNS settings.
OBJECT_DDNS_DISABLE	3	Disable the DDNS client
OBJECT_DDNS_ENABLE	4	Enable the DDNS client.

Message structure – ATTR_DDNS_GET:

Frame	Value	Description
Command	189	Command FC_SETAPPCONFIG
Method	2	Method: ATTR_DDNS_GET

Example:

Todo:

Message structure – OBJECT_DDNS_DISABLE:

Frame	Value	Description
Command	189	Command FC_SETAPPCONFIG
Method	3	Method: OBJECT_DDNS_DISABLE

Example:

Todo:

Message structure – OBJECT_DDNS_ENABLE:

Frame	Value	Description
Command	189	Command FC_SETAPPCONFIG
Method	4	Method: OBJECT_DDNS_ENABLE
Data 0	Provider	Provider: see table Provider ID
Data 1	Length	Length of hostname (Max 64 char).
Data 2...N	Hostname	Hostname string data.
Data N+1	Length	Length of username (Max 32 char)
Data N+2...O	Username	Username string data.
Data O+1	Length	Length of password (Max 32 char)
Data O+2...P	Password	Password string data.

Table – Provider ID

Value	Description
0	DDNS Client disabled.
1	NO-IP.Com provider.

Example:

Todo:

EV_APPCONFIG [15]

General:

The device replies this message to the host software in response of a FC_APPCONFIG message.

Frame	Value	Description
Command	15	Command EV_APPCONFIG
Method	Method 2-5	Method – see table.
Data 0 - N	Data	F(method) – see below

Methods

Method	Value	Description
ATTR_DDNS_GET	2	Return the current DDNS settings.

OBJECT_ DDNS_ DISABLE	3	Reply a Result code to the disable command message.
OBJECT_ DDNS_ ENABLE	4	Reply a Result code to the enable command message.

Message structure – ATTR_DDNS_GET:

Frame	Value	Description
Command	15	Command EV_APPCONFIG
Method	2	Method: ATTR_DDNS_GET
Data 0	Provider	Provider: see table Provider ID – (FC_SETAPPCONFIG message).
Data 1	Status	DDNS Status: See table
Data 2	Length	Length of hostname (Max 64 char)
Data 3...N	Hostname	Hostname string data.
Data N+1	Length	Length of Username (Max 32 char).
Data N+2...O	Username	Username string data.
Data O+1	Length	Length of password (Max 32 char)
Data O+2...P	Password	Password string data.

Table – DDNS status.

Value	Description
0	Disabled
1	Good
2	Error state
3	Wait

Example:

Todo:

Message structure – OBJECT_DDNS_DISABLE:

Frame	Value	Description
Command	15	Command EV_APPCONFIG
Method	3	Method: OBJECT_DDNS_DISABLE
Data 0	0 or 1	0 = Failed / 1= Success.

Example:

Todo:

Message structure – OBJECT_DDNS_ENABLE:

Frame	Value	Description
Command	15	Command AV_APPCONFIG
Method	4	Method: OBJECT_DDNS_ENABLE
Data 0	0 or 1	0 = Failed / 1= Success.

Example:

Todo:

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Application system mode configuration.

Overview:

Request and set the application system mode(Basic/Extended) settings.

Packet fields:

[Command , method , data , data N-1]

Messages:

FC_APPCONFIG [189]

General:

The host configuration software sends this message to the device to request and set the Application systemmode settings.

Message structure:

Frame	Value	Description
Command	189	Command FC_APPCONFIG
Method	5-6	See table Methods

Methods

Method	Value	Description
ATTR_SYSTEMMODE_GET	5	Get the current system mode settings.
ATTR_SYSTEMMODE_SET	6	Set the current system mode settings.

Message structure – ATTR_SYSTEMMODE_GET:

Frame	Value	Description
Command	189	Command FC_SETAPPCONFIG
Method	5	Method: ATTR_SYSTEMMODE_GET

Message structure – ATTR_SYSTEMMODE_SET:

Frame	Value	Description
Command	189	Command FC_SETAPPCONFIG
Method	6	Method: ATTR_SYSTEMMODE_SET
Data 0	0 - 1	0 = Basic, 1 = Extended

EV_APPCONFIG [15]

General:

The device replies this message to the host software in response of a FC_APPCONFIG message.

Frame	Value	Description
Command	15	Command EV_APPCONFIG
Method	Method 5-6	Method – see table.
Data 0 - N	Data	F(method) – see below

Methods

Method	Value	Description
ATTR_SYSTEMMODE_GET	5	Return the current system mode settings.
ATTR_SYSTEMMODE_SET	6	Reply with result code.

Message structure – ATTR_SYSTEMMODE_GET

Frame	Value	Description
Command	15	Command EV_APPCONFIG
Method	5	Method: ATTR_SYSTEMMODE_GET
Data 0	0 - 1	0 = Basic, 1 = Extended

Example:

Todo:

Message structure – ATTR_SYSTEMMODE_SET

Frame	Value	Description
Command	15	Command EV_APPCONFIG
Method	6	Method: ATTR_HOMESRVCFG_SET
Data 0	0 or 1	0 = Failed / 1= Success.

Example:

Todo:

Application DNS Provider configuration.

Overview:

Request and set the application system mode(Basic/Extended) settings.

Packet fields:

[Command , method , data , data N-1]

Messages:

FC_APPCONFIG [189]

General:

The host configuration software sends this message to the device to request and set the Application DNS provider settings.

Message structure:

Frame	Value	Description
Command	189	Command FC_APPCONFIG
Method	10-11	See table Methods

Methods

Method	Value	Description
ATTR_DNSPROVIDER_GET	10	Get the DNS provider settings.
ATTR_DNSPROVIDER_SET	11	Set the DNS provider settings.

Message structure – ATTR_DNSPROVIDER_GET:

Frame	Value	Description
Command	189	Command FC_SETAPPCONFIG
Method	10	Method: ATTR_DNSPROVIDER_GET

Message structure – ATTR_TCIPCLIENT_SET:

Frame	Value	Description
Command	189	Command FC_SETAPPCONFIG
Method	11	Method: ATTR_DNSPROVIDER_SET
Data 0	IP_Address_3	DNS Provider IP-Address – BYTE 3
Data 1	IP_Address_2	DNS Provider IP-Address – BYTE 2
Data 2	IP_Address_1	DNS Provider IP-Address – BYTE 1
Data 3	IP_Address_0	DNS Provider IP-Address – BYTE 0

EV_APPCONFIG [15]

General:

The device replies this message to the host software in response of a FC_APPCONFIG message.

Message structure:

Frame	Value	Description
Command	15	Command EV_APPCONFIG
Method	Method 10-11	Method – see table.
Data 0 - N	Data	F(method) – see below

Methods

Method	Value	Description
ATTR_DNSPROVIDER_GET	10	Reply with the DNS provider settings
ATTR_DNSPROVIDER_SET	11	Reply to the FC_SETAPPCONFIG message with a result code

Message structure – ATTR_TCIPCLIENT_GET:

Frame	Value	Description
Command	15	Command EV_APPCONFIG
Method	10	Method: ATTR_DNSPROVIDER_GET
Data 0	IP_Address_3	DNS Provider IP-Address – BYTE 3
Data 1	IP_Address_2	DNS Provider IP-Address – BYTE 2
Data 2	IP_Address_1	DNS Provider IP-Address – BYTE 1
Data 3	IP_Address_0	DNS Provider IP-Address – BYTE 0

Example:

Todo:

Message structure – ATTR_TCIPCLIENT_SET:

Frame	Value	Description
Command	15	Command EV_APPCONFIG
Method	11	Method: ATTR_DNSPROVIDER_SET
Data 0	0 or 1	0 = Failed / 1= Success.

Example:

Todo:

Application TCP/IP client configuration.

Overview:

Request and set the application tcp/ip client settings.

Packet fields:

[Command , method , data , data N-1]

Messages:

FC_APPCONFIG [189]

General:

The host configuration software sends this message to the device to request and set the Application tcp/ip client settings.

Message structure:

Frame	Value	Description
Command	189	Command FC_APPCONFIG
Method	8 - 9	See table Methods

Method	Value	Description
ATTR_TCIPCLIENT_GET	8	Get the current TCP/IP client settings.
ATTR_TCIPCLIENT_SET	9	Set the current TCP/IP client settings.

Message structure – ATTR_TCIPCLIENT_GET:

Frame	Value	Description
Command	189	Command FC_SETAPPCONFIG
Method	8	Method: ATTR_TCIPCLIENT_GET

Message structure – ATTR_TCIPCLIENT_SET:

Frame	Value	Description
Command	189	Command FC_SETAPPCONFIG
Method	9	Method: ATTR_TCIPCLIENT_SET
Data 0	IP_Address_3	IP-Address – BYTE 3
Data 1	IP_Address_2	IP-Address – BYTE 2
Data 2	IP_Address_1	IP-Address – BYTE 1
Data 3	IP_Address_0	IP-Address – BYTE 0
Data 4	IP Port	IP port MSB
Data 5	IP Port	IP port LSB
Data 6	Reserved	Reserved Byte

EV_APPCONFIG [15]

General:

The device replies this message to the host software in response of a FC_APPCONFIG message.

Message structure:

Frame	Value	Description
Command	15	Command EV_APPCONFIG
Method	Method 0-1	Method – see table.
Data 0 - N	Data	F(method) – see below

Methods

Method	Value	Description
ATTR_TCIPCLIENT_GET	8	Reply with the current TCP/IP client settings
ATTR_TCIPCLIENT_SET	9	Reply to the FC_SETAPPCONFIG message with a result code.

Message structure – ATTR_TCIPCLIENT_GET

Frame	Value	Description
Command	15	Command EV_APPCONFIG
Method	8	Method: ATTR_TCIPCLIENT_GET
Data 0	IP_Address_3	IP-Address – BYTE 3
Data 1	IP_Address_2	IP-Address – BYTE 2
Data 2	IP_Address_1	IP-Address – BYTE 1
Data 3	IP_Address_0	IP-Address – BYTE 0
Data 4	IP Port	IP port MSB
Data 5	IP Port	IP port LSB
Data 6	Status	Status Connected = 1, Status Not connected = 0

Example:

Todo:

Message structure – ATTR_TCIPCLIENT_SET

Frame	Value	Description
Command	15	Command EV_APPCONFIG
Method	9	Method: ATTR_TCIPCLIENT_SET
Data 0	0 or 1	0 = Failed / 1= Success.

Example:

Todo:

Application SNTP configuration.

Overview:

Request and set the application use of SNTP service.

Packet fields:

[Command , method , data , data N-1]

Messages:

FC_APPCONFIG [189]

General:

The host configuration software sends this message to the device to request and set the Application SNTP service settings.

Message structure:

Frame	Value	Description
Command	189	Command FC_APPCONFIG
Method	12-13	See table Methods

Methods

Method	Value	Description
ATTR_Sntp_GET	12	Get the SNTP service settings.
ATTR_Sntp_SET	13	Set the SNTP service settings.

Message structure – ATTR_Sntp_GET:

Frame	Value	Description
Command	189	Command FC_SETAPPCONFIG
Method	12	Method: ATTR_Sntp_GET

Message structure – ATTR_Sntp_SET:

Frame	Value	Description
Command	189	Command FC_SETAPPCONFIG
Method	13	Method: ATTR_Sntp_SET
Data 0	Flags	Reserved – for the moment 0
Data 1	Pool ID	Pool ID

EV_APPCONFIG [15]

General:

The device replies this message to the host software in response of a FC_APPCONFIG message.

Message structure:

Frame	Value	Description
Command	15	Command EV_APPCONFIG
Method	Method 12-13	Method – see table.
Data 0 - N	Data	F(method) – see below

Methods

Method	Value	Description
ATTR_Sntp_GET	12	Reply with the SNTP service settings
ATTR_Sntp_SET	13	Reply to the FC_SETAPPCONFIG message with a result code

Message structure – ATTR_Sntp_GET:

Frame	Value	Description
Command	15	Command EV_APPCONFIG
Method	12	Method: ATTR_Sntp_GET
Data 0	Flags	Reserved – for the moment 0
Data 1	Pool ID	Pool ID

Example:

Todo:

Message structure – ATTR_Sntp_SET:

Frame	Value	Description
Command	15	Command EV_APPCONFIG
Method	13	Method: ATTR_Sntp_SET
Data 0	0 or 1	0 = Failed / 1= Success.

Example:

Todo:

Application Time settings configuration.

Overview:

Request and set the application use of time settings.

Packet fields:

[Command , method , data , data N-1]

Messages:

FC_APPCONFIG [189]

General:

The host configuration software sends this message to the device to request and set the Application SNTP service settings.

Message structure:

Frame	Value	Description
Command	189	Command FC_APPCONFIG
Method	14-15	See table Methods

Methods

Method	Value	Description
ATTR_TIME_GET	14	Get the application time settings.
ATTR_TIME_SET	15	Set the application time settings.

Message structure – ATTR_TIME_GET:

Frame	Value	Description
Command	189	Command FC_SETAPPCONFIG
Method	14	Method: ATTR_TIME_GET

Message structure – ATTR_TIME_SET:

Frame	Value	Description
Command	189	Command FC_SETAPPCONFIG
Method	15	Method: ATTR_SNTP_SET
Data 0	Flags	Reserved
Data 1	Area	Area code
Data 2	Country	Reserved – 0 for the moment.
Data 3	UTC Zone	Signed value -12 tem +14
Data 4	DST	Daylight savings time enabled/disabled 0-1

EV_APPCONFIG [15]

General:

The device replies this message to the host software in response of a FC_APPCONFIG message.

Message structure:

Frame	Value	Description
Command	15	Command EV_APPCONFIG
Method	Method 14-15	Method – see table.
Data 0 - N	Data	F(method) – see below

Methods

Method	Value	Description
ATTR_TIME_GET	14	Reply with the application time settings
ATTR_TIME_SET	15	Reply to the FC_SETAPPCONFIG message with a result code

Message structure – ATTR_TIME_GET:

Frame	Value	Description
Command	15	Command EV_APPCONFIG
Method	14	Method: ATTR_TIME_GET
Data 0	Flags	Reserved
Data 1	Area	Area code
Data 2	Country	Reserved – 0 for the moment.
Data 3	UTC Zone	Signed value -12 tem +14
Data 4	DST	Daylight savings time enabled/disabled 0-1

Example:

Todo:

Message structure – ATTR_TIME_SET:

Frame	Value	Description
Command	15	Command EV_APPCONFIG
Method	15	Method: ATTR_TIME_SET
Data 0	0 or 1	0 = Failed / 1= Success.

Example:

Todo: